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Effects of a 30-min nap opportunity on cognitive and short-duration high-intensity performances and mood states after a partial sleep deprivation night

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ABSTRACT

This study evaluated the effects of partial-sleep-deprivation (SDN) and a 30 min nap opportunity on physical and cognitive performances and mood states. Fourteen physically active students (BMI = 232.8 ± 0.4 kg/m^2) performed the reaction time, the number cancellation (*i.e.*, assessing vigilance) and the 5-m shuttle run tests and responded to the Profile of Mood States (POMS-f) questionnaire at 18h00 after a normal-sleep (NSN) and a SDN) and after two nap conditions (Nap and no-Nap) realized between 13h00 and 13h30. Vigilance and the reaction time were better after Nap compared to no-Nap opportunity following NSN and SDN and during NSN compared to SDN only during no-Nap. Total and peak distance during the 5-m shuttle run test were higher and the fatigue index was lower during Nap compared to no-Nap condition after NSN and SDN and during NSN compared to SDN during Nap and no-Nnap. Anxiety, fatigue, confusion, and depression were lower and vigour was higher during Nap compared to no-Nap after NSN and SDN and during NSN compared to SDN during Nap and no-Nap. In conclusion, a 30-min of nap opportunity helps to overcome the negative effect of SDN on mood states as well as physical and cognitive performances.

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KEYWORDS
Sleep; nap; alertness; repeated sprint; sport

Introduction

The alternation rest-activity is a characteristic of the living world. During rest, sleep is generally considered to be a repairing process that influences the homoeostatic regulation of the autonomic, neuro-endocrine and immune nervous systems (Reilly & Edwards, 2007). In physically active persons, adequate sleep is essential for optimal muscle performance and human behaviour (Davenne, 2009; K. Trabelsi et al., 2019a).

Because of its major importance for human circadian rhythms, disruption of the sleep-wake cycle can cause adverse effects on mental and muscular performance in different settings (Davenne, 2009). In a sleep loss condition, previous studies support the hypothesis that psychomotor and cognitive performances (Daaloul et al., 2018; Fullagar et al., 2015), as well as mood states (Lastella et al., 2014), have been negatively affected. Jarraya et al. (2012, 2013) reported a negative-effect of a sleep deprivation night (SDN) on cognitive performances (*i.e.*, reaction time, attention). The reduction in cognitive performance in a disturbed sleep state could be related to the disruption of pre-frontal cortex activity due to its sensitivity to both arousal and attention (Grundgeiger et al., 2014). Also, for physical performance, previous studies reported that performance during the 30-s Wingate test was lower after SDN compared to a normal sleep nigh (NSN) (H. Chtourou et al., 2015; HajSalem et al., 2013; N. Souissi et al., 2013). Otherwise, it is important to mention that sports performance depends on the effectiveness of the physical, psychomotor and cognitive components (Mori et al., 2002) and that mood, psychomotor and cognitive functions decline is faster than the reduction in physical abilities (Davenne, 2009).

In practice, many situations that may cause disruption of sleep are cited in the literature (*e.g.*, poor sleeping conditions, anxiety or stress induced by competition and travel (Davenne, 2009; K. Trabelsi et al., 2019b, 2018)); but only a few strategies have been studied in order to overcome the effect of sleep loss or perturbation on subsequent cognitive and physical performances as well as on mood states.

One of the previously investigated strategies for improving cognitive and physical performance as well as mood states after SDN is the post-lunch short-duration naps (Brooks & Lack, 2006) and that napping can be used to reduce a number of sleep-related problems, such as impaired cognitive and physical performance (Hilditch et al., 2017; Waterhouse et al., 2007). It has been reported that naps enhance arousal levels even when the quality and the quantity of the previous nocturnal sleep are adequate (Hayashi et al., 1999; Takahashi et al., 1998). Napping is a commonly used way to overcome sleepiness and impaired performance caused by fatigue, sleep loss or extended wakefulness (Hilditch et al., 2017). In this context, it has been reported that the afternoon is the best time to take a nap because it corresponds to the peak of daytime sleepiness (Strogatz et al., 1987). Previous studies showed that during short periods of naps (*i.e.*, 5 to 15 min), the benefits are almost immediate (between 1 and 3 h) (Lovato & Lack, 2010b; Petit

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et al., 2014); while the longer periods (*i.e.*, ≥30 min) may produce sleep-inertia for a short time after waking and an improvement in cognitive performance for several hours (Hammouda et al., 2018; Lovato & Lack, 2010a; Signal et al., 2012). For example, Hammouda et al. (2018) reported that 20-min and 90-min of afternoon naps might counteract the negative effect of SDN on repeated sprints exercise. For the possible explanation of the beneficial effect of naps on cognitive and physical performance, in a recent review study, Mantua and Spencer (2017) have reported that naps, for young adults contain both non-rapid eye movement and rapid eye movement bouts. The non-rapid eye movement is associated with low energy expenditure and high neuronal synchronization. However, the rapid eye movement sleep is associated with high brain activity and energy expenditure comparable to wake. The authors concluded that a mid-day nap minimizes sleepiness, improves executive functioning and facilitates memory consolidation, subsequent learning and emotional processing.

Actually, no-firm conclusions about the effect of short nap after SDN on cognitive and physical performances as well as on mood states could be drawn from the existing data. Also, to the best of the authors' knowledge, no previous study has investigated the effect of a nap opportunity on physical performance and fatigue during the 5-m shuttle run test (5mSRT), that stimulate both the aerobic and the anaerobic pathways (6 × 30-s with 35-s of recovery in-between), after SDN. In recent studies, it has been reported that a 25 min (Abdessalem et al., 2019; Boukhris et al., 2019; Hsouna et al., 2020), 35 min and 45 min (Boukhris et al., 2019; Hsouna et al., 2019) of nap opportunities were beneficial for physical performance during the 5mSRT and the 5-jump test after NSN (see H. Chtourou et al., 2019).

Thus, the aim of the present study was to examine the effect of a 30-min nap opportunity on mood states and cognitive (*i.e.*, vigilance and reaction time) and short-duration high-intensity repetitive performance after one night of SDN and NSN. In view of the literature data, the hypothesis of the present study was that a nap opportunity of 30-min might counteract the negative effect of SDN on cognitive and short-duration high-intensity repetitive performances as well as mood states in physically active students.

Methods

Participants

Knowing the experiment procedures, fourteen healthy and physically active male students whose average age, body mass and height were 20.5 ± 1.5 years, 73.8 ± 3.3 kg, and 1.8 ± 3.5 m, respectively, voluntarily participated in the present study. The criteria for participant inclusion in this study were as follows: non-smokers, did not have pathological sleep disorders and did not consume alcohol. They regularly practise physical exercise (*e.g.*, jogging) for at least 3 hours a week. Body-mass of the participants was recorded using an impedance scale metre (Tanita, Tokyo, Japan). During the period of the study, they have standard breakfast (07h00 ± 1 h), lunch (12h00 ± 1 h) and dinner (20h00 ± 1 h) as well as sleep times (between 23h00 and 07h00 ± 1 h). The study was conducted according to the Declaration of Helsinki and the protocol was fully approved by the Research Ethics Committee before the commencement of the assessments.

Experimental design

During the week preceding the study, participants visited the laboratory on two occasions for the determination of the height and body mass as well as the familiarization with the material and the experimental procedure and took a short bout of a nap opportunity.

After four nights of habituation sessions, participants came to the laboratory to perform the experimental test session. The nocturnal sleep and nap opportunities were controlled by actigraphic recording and sleep diaries with strict supervision by experimenters and cameras during all test sessions. Using a within-subjects design, during the present study, participants performed four test sessions separated by four days of rest: two sleep conditions (NSN and SDN) and two nap opportunity conditions (with and without nap). The 14 participants were divided into four groups: two groups of four participants and two groups of three participants. At the beginning of the experimental protocol, all groups were of four participants; but, due to injuries, two participants could not complete the experimental protocol. The test sessions were conducted at 18h00 in a counterbalanced order:

- For the first group of four participants: NSN with nap opportunity, SDN without nap opportunity, NSN without nap opportunity and SDN with nap opportunity.
- For the second group of three participants: SDN without nap opportunity, NSN without nap opportunity, SDN with nap opportunity and NSN with nap opportunity.
- For the third group of four participants: NSN without nap opportunity, SDN with nap opportunity, NSN with nap opportunity and SDN without nap opportunity.
- For the fourth group of three participants: SDN with nap opportunity, NSN with nap opportunity, SDN without nap opportunity and NSN without nap opportunity.

The nap was realized between 13h00 and 13h30 with a duration of 30-min. During the NSN, participants slept from 22h00 to 06h00. During the SDN, participants slept from 22h00 to 03h00.

To keep the participants awake during SDN, the no-nap condition and during the free time they were allowed to do passive activities (*e.g.*, use computers, watching television, playing card or video-games, reading, listen to the music). Each test session consisted of the accomplishment of the following tests: the mood states questionnaire (Profile Mood State Questionnaire (POMS-f)) and the digit cancellation (*i.e.*, for the assessment of vigilance) and the reaction time tests. Subsequently, after a 15-min warm-up, participants completed the 5mSRT.

Also, blood samples were drawn from an antecubital vein into 2 mL serum vacutainer tubes before and 3-min after the 5mSRT. Samples were centrifuged at 3000 rpm for 10 min and serum was separated from blood cells and stored at -20°C until analysed. Plasma lactate assay was analysed using a kit supplied by the company Cobas (kit Cobas c400).

All test sessions were conducted during the month of April under similar conditions of temperature (~27°C) and humidity (~43%). Participants were asked to maintain their usual physical activity level during the period of the study and to avoid any intense activity during the day preceding each test session.

5-m shuttle run test

The 5mSRT was performed in a gymnasium. Six beacons were placed 5 m apart in a straight line to cover a total distance of 25 m. As described by Boukhris et al. (2018), the participants were instructed to perform six repetitions of maximal sprints during 30-s with a recovery period of 35-s in-between. They began the test in line with the first beacon, they sprinted 5 m to a second beacon and returned back to the first beacon, then they sprinted 10 m to a third beacon and returned back to the first beacon, etc., until the end of the 30-s. The distance covered by each participant was recorded to the nearest of 1-m during each 30-s shuttle.

The following parameters were then calculated (Boddington et al., 2001):

- Peak distance (PD) = the longest distance covered during a 30-s shuttle;
- Total distance (TD) = the total distance covered during the 6 × 30-s shuttles;
- Fatigue index (FI) = calculated according to the following equation:

$$FI\ (\%) = [(\text{Shuttle } 1 - \text{Shuttle } 6)/\text{Shuttle } 1] \times 100$$

Profile of Mood States (POMS-f)

The state of the mood was assessed through the POMS-f questionnaire (Cayrou et al., 2003). The POMS-f consists of 65 adjectives developed to measure seven aspects of mood (anxiety/tension, depression/dejection, anger/hostility, confusion/aberration, vigour/activity, fatigue/inertia and friendship). The answers to each item belonging to a range of 0 to 4 (0 indicates "No of all" and 4 indicates "Extremely").

The digit cancellation test

The digit cancellation test is composed of four pages (Hatta et al., 2012). Each page contains 600 numbers of 1 to 5 digits arranged in 36 lines. The test consists of the cancellation of the maximum 3-digits numbers during 1-min. The number of targets is 187 with 2 to 8 numbers of 3-digits randomly allocated per line. The numbers are, separated by a point, preceded and followed, by a space. One point is awarded, for each of the 187, correctly detected targets. The final score represented the vigilance of the participant (Hatta et al., 2012).

Simple reaction time

The reaction time was used as an index of processing speed. As in the study of N. Souissi et al. (2013), the evaluation of the simple reaction time was carried out using the "React" software. The test consisted of responding, as quickly as possible, to a visual stimulus by pressing a key on a microcomputer. Ten attempts were realized and the mean value has been calculated for each participant.

Statistical analysis

Statistical tests were processed using STATISTICA Software (StatSoft, France). Values are expressed as mean ± standard deviation (M ± SD). As the sphericity assumption was not violated and after normality confirmation by the Shapiro–Wilk test, the results (*i.e.*, except for blood lactate) were analysed using a two-factor repeated measures analysis of variance (ANOVA) (2 [Nap] × 2 [Sleep]). For blood lactate, a three-factor ANOVA was utilized (3 [Exercise] × 2 [Nap] × 2 [Sleep]). Effect sizes were reported as partial eta-squared [(η_p^2): $0.01 \le \eta_p^2 \le 0.059$ = small; $0.06 \le \eta_p^2 \le 0.139$ = medium; $0.14 \le \eta_p^2$ = large] and Cohen's d that was calculated to determine the magnitude and practical relevance of changes [< 0.2 = trivial, 0.2–0.6 = small, 0.6–1.2 = moderate, 1.2–2.0 = large, and > 2.0 = very large (Hopkins, 2002)].

When appropriate, significant differences between means were tested using the Bonferroni post hoc test adjusted p-value. The level of statistical significance was set at $p < 0.05$. Exact corrected *p*-values have been reported; results given as "0.000" in the statistics output have been reported as "<0.0005".

Results

Cognitive and physical performances

The results of vigilance, reaction time, total distance, peak distance and fatigue index recorded after NSN and SDN during the nap and the no-nap opportunity conditions are presented in Figure 1. The results of the ANOVA analysis are presented in Table 1.

Vigilance

The statistical analysis showed significant main effects of sleep and nap and a significant interaction sleep × nap.

The post-hoc test revealed that vigilance was better during the nap compared to the no-nap condition after NSN ($p < 0.0005$; d = 17.1 (very large)) and SDN ($p < 0.0005$; d = 1.52 (large)). Also, vigilance was better during NSN compared to SDN only during the no-nap condition ($p < 0.0005$; d = 16.83 (very large)) (Figure 1(a)).

Simple reaction time

The statistical analysis showed significant main effects of sleep and nap. However, the interaction sleep × nap was not significant.

The post-hoc test revealed that the simple reaction time was better during the nap compared to the no-nap condition after NSN ($p = 0.0008$; d = 1.26 (large)) and SDN ($p < 0.0005$; d = 2.11 (very large)). Also, the simple reaction time was better during NSN compared to SDN only during the no-nap condition ($p = 0.002$; d = 1.27 (large)) (Figure 1(b)).

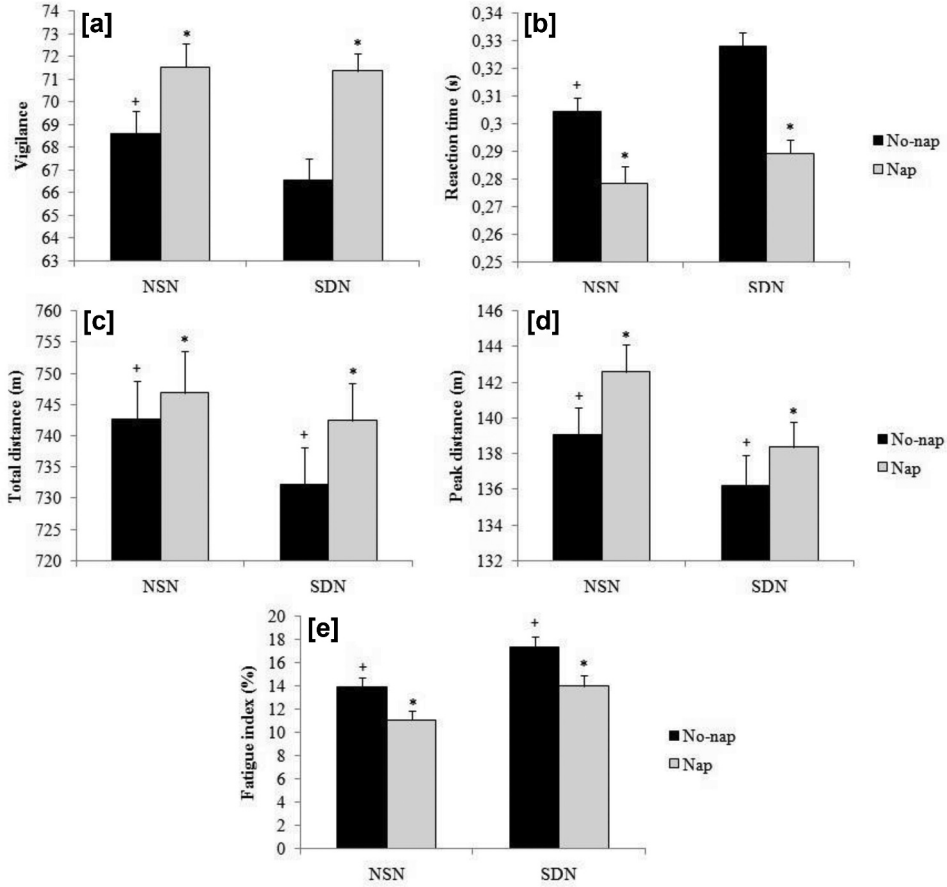

Figure 1. Vigilance [a], reaction time [b], total distance [c], peak distance [d] and fatigue index [e] recorded after the normal sleep night (NSN) and the sleep deprivation night (SDN). *: Significant difference compared to no-nap condition; +: Significant difference compared to SDN.

Table 1. ANOVA results for the cognitive and physical parameters and mood states.

	Nap				Sleep				Sleep × Nap			
	F	Degrees of freedom	η_p^2	p	F	Degrees of freedom	η_p^2	p	F	Degrees of freedom	η_p^2	p
Vigilance	261.1	1,13	0.95	<0.0005	11.6	1,13	0.47	0.0046	13.2	1,13	0.50	0.0030
Simple reaction time	295.8	1,13	0.95	<0.0005	25.6	1,13	0.66	<0.0005	3.5	1,13	0.21	0.0822
Total distance during the 5mSRT	80.6	1,13	0.86	<0.0005	50.4	1,13	0.80	<0.0005	16.8	1,13	0.56	0.0012
Peak distance during the 5mSRT	21.6	1,13	0.62	<0.0005	54.1	1,13	0.81	<0.0005	2.9	1,13	0.18	0.1118
Fatigue index during the 5mSRT	80.6	1,13	0.86	<0.0005	50.4	1,13	0.79	<0.0005	16.7	1,13	0.56	<0.0005
Anxiety	114.4	1,13	0.89	<0.0005	83.2	1,13	0.86	<0.0005	10.5	1,13	0.44	0.0064
Fatigue	134.0	1,13	0.91	<0.0005	118.3	1,13	0.90	<0.0005	0.5	1,13	0.03	0.5107
Confusion	107.8	1,13	0.89	<0.0005	99.1	1,13	0.88	<0.0005	7.3	1,13	0.36	0.0180
Depression	189.9	1,13	0.89	<0.0005	40.4	1,13	0.76	<0.0005	2.2	1,13	0.14	0.1648
Vigour	15.3	1,13	0.54	0.0017	11.0	1,13	0.46	0.0055	14.2	1,13	0.52	00.23

Total distance during the 5mSRT

The statistical analysis showed significant main effects of sleep and nap. Likewise, the interaction sleep × nap was significant.

The post hoc test revealed that the total distance was higher during the nap compared to the no-nap condition after NSN ($p = 0.01$; $d = 0.17$ (trivial)) and SDN ($p < 0.0005$; $d = 0.45$ (small)). Also, the total distance was lower during SDN compared to NSN during the no-nap ($p < 0.0005$; $d = 0.46$ (small)) and the nap ($p = 0.006$; $d = 0.18$ (trivial)) conditions (Figure 1(c)).

Peak distance during the 5mSRT

The statistical analysis showed significant main effects of sleep and nap. However, the interaction sleep × nap was not significant.

The post hoc test revealed that peak distance was higher during the nap compared to the no-nap condition after NSN ($p < 0.0005$; $d = 0.6$ (moderate)) and SDN ($p = 0.013$; $d = 0.36$ (small)). Also, peak distance was lower during SDN compared to NSN during the no-nap ($p < 0.0005$; $d = 0.46$ (small)) and the nap ($p < 0.0005$; $d = 0.75$ (moderate)) conditions (Figure 1(d)).

The fatigue index during the 5mSRT

The statistical analysis showed significant main effects of sleep and nap. Also, the interaction sleep × nap was significant.

The post hoc test revealed that FI was lower during the nap compared to the no-nap condition both after NSN ($p = 0.003$; $d = 4.65$ (very large)) and SDN ($p = 0.0006$; $d = 1.03$ (moderate)). FI was higher after SDN compared to NSN during both the no-nap ($p = 0.0005$; $d = 1.11$ (moderate)) and the nap conditions ($p = 0.002$; $d = 4.24$ (very large)) (Figure 1(e)).

Blood lactate

Blood lactate concentrations measured before, just after and 3-min after the 5mSRT after the NSN and SDN are presented in Figure 2 and the results of the ANOVA analysis are presented in Table 2.

The statistical analysis showed significant main effects for exercise, sleep and nap and significant interactions nap × sleep and exercise × nap × sleep. However, the interactions exercise × sleep and exercise × nap were not significant.

During NSN and SDN and the nap and the no-nap conditions, the post hoc test revealed that blood lactate was significantly higher at 3-min after the 5mSRT compared to before ($p < 0.0005$; $d > 2.0$ (very large)) and after ($p < 0.0005$; $d > 2.0$ (very large)) the 5-mSRT and after the 5mSRT compared to before the 5mSRT ($p < 0.0005$; $d > 2.0$ (very large)).

During the measure realized after the 5-mSRT, blood lactate was higher after SDN compared to NSN for the no-nap condition ($p < 0.0005$; $d = 0.64$ (moderate)). Also, after SDN, blood lactate was significantly higher after the no-nap compared to the nap condition ($p = 0.0009$; $d = 0.61$ (moderate)).

Mood States

Mood state parameters recorded after NSN and SDN during the nap and the no-nap opportunity conditions are presented in Table 3 and the results of the ANOVA are presented in Table 1.

The statistical analysis revealed a significant main effect of sleep for anxiety, fatigue, confusion, depression and vigour. Likewise, a significant main effect of nap for anxiety, fatigue, confusion, depression and vigour was observed. The interaction sleep × nap was significant for anxiety, confusion and vigour. However, no-significant interaction sleep × nap was observed for fatigue and depression (Table 3).

The post-hoc test revealed that anxiety score was significantly lower during the nap compared to the no-nap condition after NSN ($p = 0.007$; $d = 1.04$ (moderate)) and SDN ($p = 0.001$; $d = 3.49$ (very large)). Also, anxiety score was higher during SDN compared to NSN during the no-nap ($p = 0.001$; $d = 2.47$ (very large)) and the nap ($p = 0.006$; $d = 2.19$ (very large)) conditions. The fatigue score was significantly lower during the nap compared to the no-nap condition after NSN ($p = 0.0008$; $d = 1.99$ (large)) and SDN ($p = 0.004$; $d = 1.74$ (large)). Also, fatigue score was higher during SDN compared to NSN during the no-nap ($p = 0.016$; $d = 1.44$ (large)) and the nap ($p = 0.002$; $d = 1.57$ (large)) conditions. The confusion score was significantly lower during the nap compared to the no-nap condition after NSN ($p = 0.0005$; $d = 1.38$ (large)) and SDN ($p < 0.0005$; $d = 2.35$ (very large)). Also, confusion score was higher during SDN compared to NSN during the no-nap ($p < 0.0005$; $d = 2.57$ (very large)) and the nap ($p < 0.0005$; $d = 1.63$ (large)) conditions. The depression score was lower during the nap compared to the no-nap condition after NSN ($p < 0.0005$; $d = 1.63$ (large)) and SDN ($p < 0.0005$; $d = 9.88$ (very large)). Also, depression score was higher during SDN compared to NSN during the no-nap ($p < 0.0005$; $d = 8.43$ (very large)) and the nap ($p < 0.0005$; $d = 2.23$ (very large)) conditions. The vigour score was higher during the nap compared to the no-nap condition only during SDN ($p = 0.0012$; $d = 1.6$ (large)). The vigour score was lower during SDN compared to NSN only during the no-nap condition ($p = 0.0012$; $d = 1.47$ (large)).

Discussion

The aim of the present study was to investigate the effect of a 30-min nap opportunity on cognitive (*i.e.*, vigilance and reaction time) and short-duration high-intensity repetitive performance

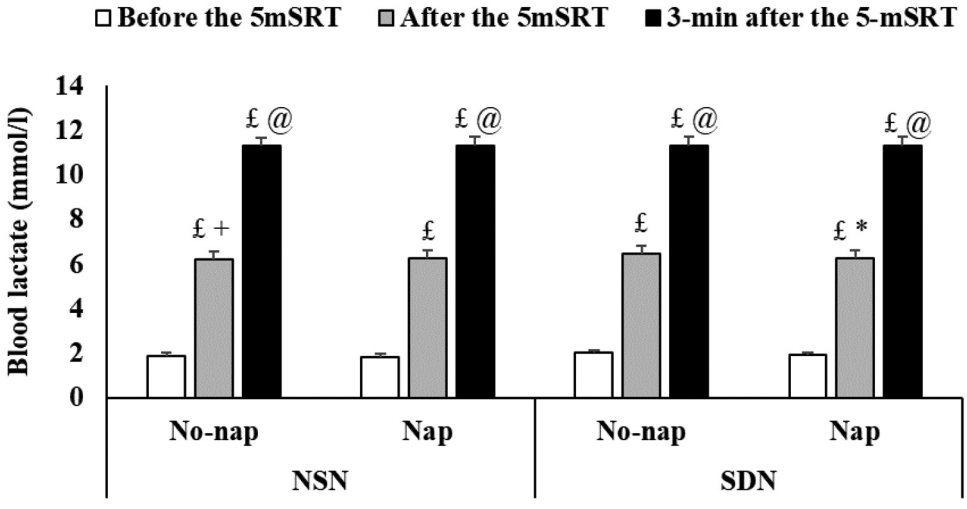

Figure 2. Blood lactate measured before, just after and 3-min after the 5-m shuttle run test (5mSRT) after the normal sleep night (NSN) and the sleep deprivation night (SDN). *: Significant difference compared to no-nap condition; +: Significant difference compared to SDN; £: Significant difference compared to before the 5mSRT; @: Significant difference compared to after the 5mSRT.

Table 2. ANOVA results for blood lactate.

	F	Degrees of freedom	η_p^2	p
Exercise	4636.9	2,26	0.99	<0.0005
Sleep	21.6	1,13	0.62	<0.0005
Nap	12.3	1,13	0.48	0.0038
Exercise × Sleep	2.79	2,26	0.17	0.0796
Exercise × Nap	3.33	2,26	0.20	0.0516
Nap × Sleep	10.75	1,13	0.45	0.0059
Exercise × Nap × Sleep	3.84	2,26	2.22	0.0344

Table 3. Mood states recorded after the normal sleep night (NSN) and the sleep deprivation night (SDN).

	NSN		SDN	
	No-nap	Nap	No-nap	Nap
Anxiety (AU)	8.71 ± 2.20+	6.64 ± 1.74*+	13.71 ± 1.70	9.36 ± 2.30*
Fatigue (AU)	10.14 ± 1.61+	7.36 ± 1.15*+	12.07 ± 1.00	9.78 ± 1.58*
Confusion (AU)	9.64 ± 1.78+	7.36 ± 1.60*+	13.93 ± 1.54	10.07 ± 1.73*
Depression (AU)	11.21 ± 1.37+	9.64 ± 1.01*+	13.50 ± 0.94	11.36 ± 1.08*
Vigour (AU)	14.86 ± 2.57+	14.71 ± 2.30	11.93 ± 1.14	14.86 ± 2.32*

*: significant difference compared to no-nap condition; +: significant difference compared to SDN.

as well as on psychological (*i.e.*, mood states) and physiological (*i.e.*, blood lactate) parameters following SDN and NSN. The main findings of the present study were that (*i*) SDN negatively affects short-duration exercise and cognitive performances as well as mood states, (*ii*) a nap opportunity of 30-min has a beneficial effect on cognitive performance and mood states and improves short-duration exercise performance, and (*iii*) a nap opportunity of 30-min helps to overcome the negative effect of SDN on short-duration exercise and cognitive performances as well as on mood states.

The alteration of cognitive and physical performances after SDN compared to NSN is in agreement with several previous studies (Baati et al., 2015; H. Chtourou et al., 2016, 2015). In this context, for physical performance, Abedelmalek et al. (2013), HajSalem et al. (2013), N. Souissi et al. (2013), and H. Chtourou et al. (2015) reported that peak and mean power during the 30-s Wingate test were significantly lower during SDN compared to NSN. After a total sleep night, Baati et al. (2015) reported a significant decrease in peak power during the 10 × 6-s repeated maximal cycling exercise. Also, these studies (Abedelmalek et al., 2013; H. Chtourou et al., 2015; HajSalem et al., 2013; N. Souissi et al., 2013) about physical exercise reported a significant increase in the fatigue index recorded during the 30-s Wingate test after SDN compared to NSN. Moreover, in agreement with the results of the present study, several previous studies reported a significant decline in cognitive performance after SDN compared to NSN (Baati et al., 2020, 2015; Jarraya et al., 2012, 2013; M. Souissi et al., 2014). Jarraya et al. (2012, 2013) reported that vigilance recorded by the number cancellation test and the reaction time was negatively affected after SDN compared to NSN.

The present study' results showed an alteration in mood states (*i.e.*, an increase in anxiety, fatigue, confusion and depression and a reduction in vigour) without changes in blood lactate after SDN compared to NSN. In agreement, Scott et al. (2006) reported a significant increase in negative mood sensations after SDN compared to NSN. The same results are reported by M. Souissi et al. (2014) showing a significant increase in negative mood states after SDN compared to NSN. For physiological responses, although it has been reported that blood lactate was higher during submaximal exercise performed after both a delayed bedtime and an early awakening (Mougin et al., 2001), the present study' results showed no significant changes in blood lactate after SDN compared to NSN.

The results of the present study showed that a 30-min nap opportunity improved cognitive and physical performances as well as mood states both after SDN and NSN. However, a better improvement was observed after SDN. This better improvement might help to overcome the impaired performances caused by fatigue, sleep loss or extended wakefulness. These observations are in agreement with those of Waterhouse et al. (2007) who showed that 30-min of nap resulted in a significant increase in performance during a sprint exercise. Also, Tietzel and Lack (2001) showed an improvement in vigilance and cognitive performance after a nap of 10 min. Recently, it has been reported that a 25-min (Abdessalem et al., 2019; Boukhris et al., 2019; Hsouna et al., 2020), 35-min and 45-min (Boukhris et al., 2019; H. Chtourou et al., 2019) of nap opportunities improved the total and the peak distances during the 5mSRT after NSN. After SDN, Lahl et al. (2008) and Gillberg et al. (1996) found an improvement in vigilance following a 30-min nap. Nishida and Walker (2007) reported that short naps could counteract the decline in performance during a short-duration maximal exercise induced by a disturbed sleep night. The authors suggested that this positive effect is not simply associated with a particular sleep stage, but with unique electrophysiological events at anatomically discrete locations of the human brain that has facilitated motor memories dynamically during naps. Similarly, Hammouda et al. (2018) reported an improvement in performance during repeated sprint exercise after napping following SDN.

In the other hand, the nap duration is important for the explanation of its beneficial effect on performance. Indeed, less than 20 min is considered optimal to avoid slow-wave sleep (deep sleep) (Petit et al., 2014). Alternatively, 90 min is also considered optimal as this allows a complete sleep cycle (non-rapid eye movements (non-REM) and rapid eye movements (REM)) to occur, reducing the effects of sleep inertia which is a temporary reduction in arousal and performance associated with slow-wave sleep in NREM (Van Dongen & Dinges, 2005). Naps are likely to occur in the mid-afternoon, when there is a dip in the circadian rhythm following lunch, which is a period of time when attention and alertness are reduced (Milner & Cote, 2009).

The improvement of cognitive and physical performances after a 30-min nap opportunity could be partially explained by the fact that this period could be comparable to a night's sleep and that this period contains both non-rapid eye movement and rapid eye movement bouts (Mantua & Spencer, 2017). Mantua and Spencer (2017) concluded that these sleep phases help to minimize sleepiness, improve executive functioning and facilitate memory consolidation, subsequent learning and emotional processing. Also, the reduction of anxiety, fatigue, confusion and depression and the improvement of vigour could explain the positive effects of the 30-min nap opportunity on cognitive and

physical performances. These findings are consistent with previous studies indicating that short naps produce a wide range of benefits from increasing alertness to improving cognitive outcomes (Daaloul et al., 2018; Milner & Cote, 2009). Evidently, optimal alertness levels and cognitive skills are integral parts of the abilities required to reach the highest elite-sport performances (Daaloul et al., 2018; Milner & Cote, 2009). In agreement, Van Dongen and Dinges (2005) reported that naps reduced both anxiety and fatigue and increased vigour. Likewise, Lovato and Lack (2010a, 2010b) showed that naps induced a reduction in drowsiness and improved cognitive performance. Also, Takahashi et al. (1998) and Hayashi et al. (1999) revealed an improvement in alertness and reaction time following a nap after an NSN. Sports performances depend on the complexity and effectiveness of the physical, psychomotor and cognitive components necessary for the realization of a specific sporting task (Mori et al., 2002). For physiological parameters, the present study' results showed no significant changes after napping during SDN and NSN.

Limitations, strength and practical applications

A limitation of the present study was the applied protocols of the sleep deprivation (*i.e.*, the obligation to be awake at 03h00) and the nap (*i.e.*, a 30-min nap opportunity at 13h00), which might differ from those used in other studies. Furthermore, the present study utilized only one night of partial sleep restriction and we suggest that the reduction in both cognitive and physical performances could be higher after 2–3 weeks of sleep deprivation. Thus, further studies could examine the effect of repetitive nights of sleep deprivation on physical and cognitive performance. Also, caution would be needed to generalize the role of the sleep deprivation and the nap on psychological parameters and physiological performances in other settings. Indeed, although the results could suggest that amateur and elites athletes could benefit from the nap opportunity, the conclusions of the present study are specific to healthy and physically active male students. On the other hand, strength of the study was its novelty as it was the first to examine the combined effect of the sleep deprivation and the nap opportunity on both physiological and psychological measures. Although individual differences play an important role with respect to the potential benefits of napping opportunity on performance, considering the role of the sleep deprivation and the nap on psychological parameters and physiological performances of a wide range of activities (*e.g.*, sport, military) (Keramidas et al., 2018), these findings would have great practical applications.

Conclusions

In conclusion, the results of the present study revealed that a 30-min nap opportunity improved mood states as well as cognitive and physical performances after a sleep deprivation night and after a normal sleep night. However, no-significant change was observed in physiological parameters in response to exercise. Moreover, the results could be important for athletes and coaches for better recovery as napping opportunity is a beneficial strategy for enhancing cognitive and physical performances among athletes during daily training or before athletic events and counteracting the deleterious effects of partial sleep deprivation caused by poor sleeping conditions, anxiety or stress induced by competition.

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Conflict of interest

The authors have no known conflict of interest with this study or its results and do not endorse any commercially available product used in this investigation.

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